cp_9b

1

Print Numbers Divisible by T | 30/30

Easy

Write a program that reads two numbers `N` and `T`, and prints all the numbers from **1** to `N` that are divisible by `T`.

Input

The first line of input contains an integer representing `N`.
The second line of input contains an integer representing `T`.

Output

The output should contain all the numbers from **1** to `N` that are divisible by `T`, each on a new line.

Explanation

For example, if the given numbers are `N = 10` and `T = 3`,

- Numbers from **1** to **10** are **1**, **2**, **3**, **4**, **5**, **6**, **7**, **8**, **9**, and **10**.
- Numbers from **1** to **10** that are divisible by **3** are **3**, **6** and **9**.

The output should be,

```
3
6
9
```

code

```
n=int(input())
t=int(input())
for i in range(1,n+1):
    rem=i%t
    if rem==0:
        print(i)
```

2

Sum of Numbers Divisible by T | 35/35

Given three integers, write a program to print the sum of the numbers divisible by the given number T from M to N.

Input

The first line of input will contain an integer (T).
The second line of input will contain an integer (M).
The third line of input will contain an integer (N).

Output

The output should be a single line containing the sum of the numbers divisible by the given number T.

Explanation

For example, if the given number T is 2, M is 5, and N is 9.
The numbers which are in the range from 5 to 9 and divisible by 2 are 6 and 8.
So the output should be the sum of these numbers, which is 14.

Sample Input 1

```
2
5
9
```

Sample Output 1

```
14
```

code

```
t=int(input())
m=int(input())
n=int(input())
sum=0
for i in range(m,n+1):
    rem=i%t
    if rem==0:
        sum=sum+i
print(sum)
```

3

Count of Numbers Divisible by T | 35/35

Easy

Write a program that reads two numbers `N` and `T` and prints the count of numbers from **1** to `N` that are divisible by `T` .

Input

The first line of input contains an integer representing `N` .
The second line of input contains an integer representing `T` .

Output

The output should be a single line containing an integer that is the count of numbers from **1** to `N` that are divisible by `T` .

Explanation

For example, if the given numbers are `N = 12` and `T = 3` ,

- Numbers from **1** to **12** are **1**, **2**, **3**, **4**, **5**, **6**, **7**, **8**, **9**, **10**, **11**, and **12**.
- Numbers from **1** to **12** that are divisible by **3** are **3**, **6**, **9**, and **12**.
- The count of numbers from **1** to **12** that are divisible by **3** is **4**.

The output should be **4**.

code

```
n=int(input())
t=int(input())
count=0
for i in range(1,n+1):
    rem=i%t
    if rem==0:
        count=count+1
print(count)
```

4

Print Numbers Divisible by 2 and 3 | 35/35

Easy

Write a program that reads a number `N` and prints all the numbers from **1** to `N` that are divisible by both **2** and **3**.

Input

The input will be a single line containing an integer representing `N` .

Output

The output should contain all the integers from **1** to `N` that are divisible by both **2** and **3**, each on a new line.

Explanation

For example, if the given number is `N = 15` ,

- Numbers from **1** to **15** are **1**, **2**, **3**, **4**, **5**, **6**, **7**, **8**, **9**, **10**, **11**, **12**, **13**, **14**, and **15**.
- Numbers from **1** to **15** that are divisible by both **2** and **3** are **6** and **12**.

The output should be,

```
6
12
```

code

```
n=int(input())
for i in range(1,n+1):
    rem1=i%2
    rem2=i%3
    if rem1==0 and rem2==0:
        print(i)
```

5

Count of Numbers Divisible by 2 and 3 | 30/30

Easy

Write a program that reads a number `N` and prints the count of numbers from **1** to `N` that are divisible by both **2** and **3**.

Input

The input will be a single line containing an integer representing `N` .

Output

The output should be a single line containing an integer that is the count of numbers from **1** to `N` that are divisible by both **2** and **3**.

Explanation

For example, if the given number is `N = 12` ,

- Numbers from **1** to **12** are **1**, **2**, **3**, **4**, **5**, **6**, **7**, **8**, **9**, **10**, **11** and **12**.
- Numbers from **1** to **12** that are divisible by both **2** and **3** are **6** and **12**.
- The count of numbers from **1** to **12** that are divisible by both **2** and **3** is **2**.

The output should be **2**.

code

```
n=int(input())
count=0
for i in range(1,n+1):
    rem1=i%2
    rem2=i%3
    if rem1==0 and rem2==0:
        count=count+1
print(count)
```

6

Print Required Characters | 30/30

Easy

Write a program that reads a string and prints the **a** and **z** characters in the string.

Input

The input will be a single line containing a string.

Output

The output should contain all the **a** and **z** characters in the string, each on a new line.

Explanation

For example, if the given string is **zigzag**,

- The characters in the string **zigzag** are **z**, **i**, **g**, **z**, **a** and **g**.
- The **a** and **z** characters in the string are **zza**.

The output should be,

```
z
z
a
```

code

```
st=input()
len_st=len(st)
for i in range(len_st):
    if st[i]=="z":
        print("z")
    elif st[i]=="a":
        print("a")
```

7

Print Vowels | 30/30

Easy

Write a program that reads a string and prints the vowels in the string.

Note

Vowels are **a**, **e**, **i**, **o**, and **u**.

Input

The input will be a single line containing a string.

Output

The output should contain all the vowels in the string, each on a new line.

Explanation

For example, if the given string is **indian**,

- The vowels in the string **indian** are **i**, **i**, and **a**.

The output should be,

```
i
i
a
```

code

```
st=input()
len_st=len(st)
for i in range(len_st):
    if st[i]=="a":
        print("a")
    elif st[i]=="e":
        print("e")
    elif st[i]=="i":
        print("i")
    elif st[i]=="o":
```

```
        print("o")
    elif st[i]=="u":
        print("u")
```

8

Count of Vowels | 35/35

Easy

Write a program that reads a string and prints the count of vowels in the string.

Note

Vowels are **a**, **e**, **i**, **o**, and **u**.

Input

The input will be a single line containing a string.

Output

The output should be a single line containing an integer that is the count of vowels in the string.

Explanation

For example, if the given string is **code**,

- The characters of the string **code** are **c**, **o**, **d** and **e**.
- The vowels in the string **code** are **o** and **e**.
- The count of vowels in the string **code** is **2**.

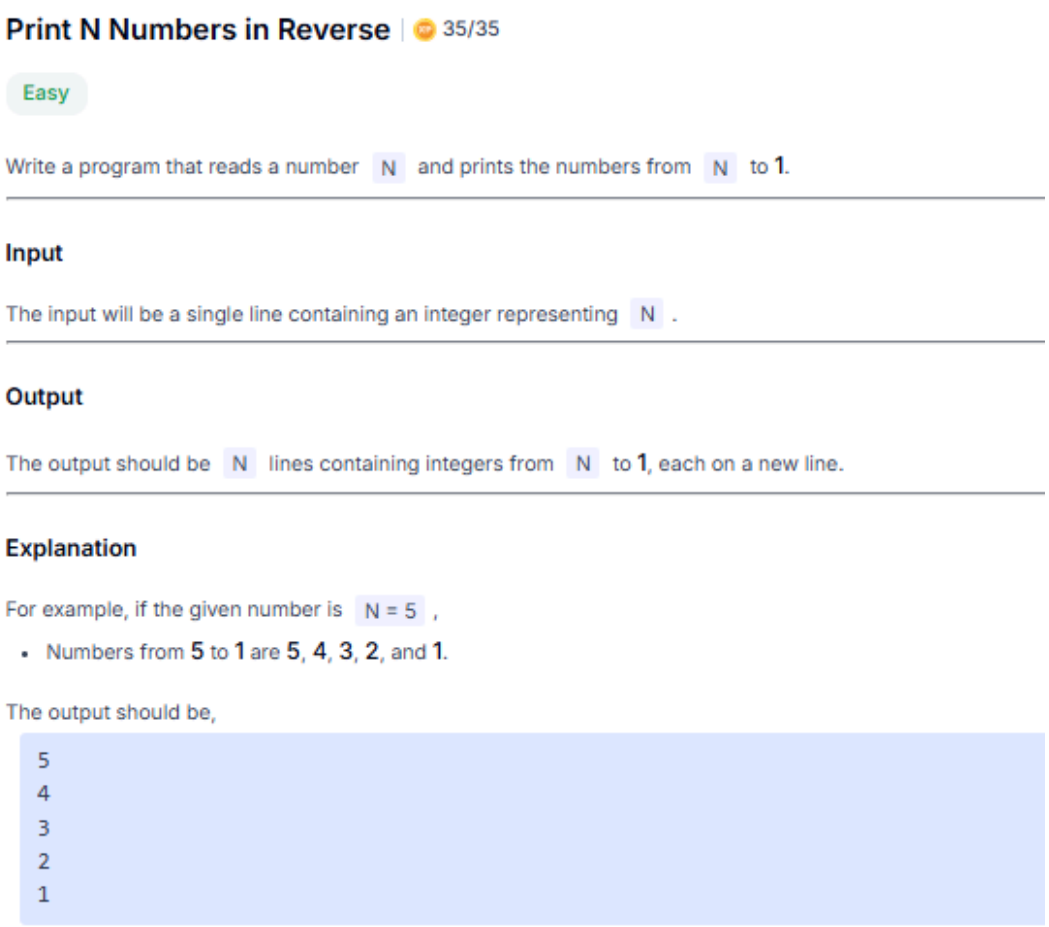
The output should be **2**.

code

```
st=input()
len_st=len(st)
count=0
for i in range(len_st):
```

```
    if st[i]in ["a","e","i","o","u"]:
        count=count+1
print(count)
```

9



Print N Numbers in Reverse | 35/35

Easy

Write a program that reads a number `N` and prints the numbers from `N` to **1**.

Input

The input will be a single line containing an integer representing `N` .

Output

The output should be `N` lines containing integers from `N` to **1**, each on a new line.

Explanation

For example, if the given number is `N = 5` ,

- Numbers from **5** to **1** are **5**, **4**, **3**, **2**, and **1**.

The output should be,

```
5
4
3
2
1
```

code:

```
n=int(input())
for i in range(1,n+1):
    print(n)
    n=n-1
```

10

Print from M to N in Reverse | 35/35

Easy

Write a program that reads two numbers `M` and `N`, and prints the numbers from `N` to `M`.

Input

The first line of input contains an integer representing `M`.
The second line of input contains an integer representing `N`.

Output

The output should contain all the integers from `N` to `M`, each on a new line.

Constraints

`N` should be greater than or equal to `M`.

Explanation

For example, if the given numbers are `M = 2` and `N = 5`,

- The numbers from **5** to **2** are **5**, **4**, **3** and **2**.

The output should be,

```
5
4
3
2
```

code

```
m=int(input())
n=int(input())
for i in range (m,n+1):
    print(n)
    n=n-1
```

11

Print Characters in Reverse | 30/30

Easy

Write a program that reads a string and prints the characters of a string in reverse order.

Input

The input will be a single line containing a string.

Output

The output should contain all the characters of the string in reverse order, each on a new line.

Explanation

For example, if the given string is **scale**,

- The characters in the string **scale** are **s**, **c**, **a**, **l**, and **e**.
- The characters in reverse order are **e**, **l**, **a**, **c**, and **s**.

The output should be,

```
e
l
a
c
s
```

code

```
s=input()
i_=len(s)-1
for i in range(i_,-1,-1):
    print(s[i])
```

12

Print Odd Numbers from N to M | 40/40

Easy

Write a program that reads two numbers `M` and `N`, and prints the odd numbers from `N` to `M` separated by a space.

Input

The first line of input contains an integer representing `M`.
The second line of input contains an integer representing `N`.

Output

The output should be a single line containing a string that has space-separated odd numbers from `N` to `M`, as shown in the sample output.

Constraints

`N` is always greater than or equal to `M`.

Explanation

For example, if the given numbers are `M = 1` and `N = 10`,

- Numbers from **10** to **1** are **10**, **9**, **8**, **7**, **6**, **5**, **4**, **3**, **2**, and **1**.
- The odd numbers from **10** to **1** are **9**, **7**, **5**, **3**, and **1**.

The output should be,

```
9 7 5 3 1
```

code

```
m=int(input())
n=int(input())
st=""
for i in range(m,n+1):
    rem=i%2
    if rem==1:
        str_i=str(i)+" "
        st=str_i+st
print(st)
```

13

Inverted Right Angled Triangle | 30/30

Write a program that reads a number `N` and prints an **Inverted Right Angled Triangle** of `N` rows using stars (`*`).

Note

There is a space after every star.

Input

The input will be a single line containing an integer representing `N` .

Output

The output should be `N` rows containing stars (`*`) as an **Inverted Right Angled Triangle** shown in the sample output.

Explanation

For example, if the given number is `N = 5` ,

- Print the stars (`*`) on **5** rows as shown in the given image.

code

```
n=int(input())
for i in range(1,n+1):
    row="* "*n
    n=n-1
    print(row)
```

14

Inverted Right Angled Triangle - 2 | 40/40

Easy

Write a program that reads a number `N` and prints an **Inverted Right Angled Triangle** of `N` rows using numbers.

```
4 4 4 4
3 3 3
2 2
1
```

Note

There is a space after every number.

Input

The input will be a single line containing an integer representing `N` .

Output

The output should be `N` rows containing numbers as an **Inverted Right Angled Triangle** shown in the sample output.

Explanation

For example, if the given number is `N = 4` ,

- Print the numbers on **4** rows as shown in the given image.

Note

In the given image, there is a space after every number.

The output should be,

```
4 4 4 4
3 3 3
2 2
1
```

code

```
n=int(input())
for i in range(1,n+1):
    str_n=str(n)+" "
    row=str_n*n
```

```
    n=n-1
    print(row)
```